

TRANSLATING WORD RIDDLES TO MATH CODES



ISHAAN'S AUTORICKSHAW RIDDLE

Ishaan spots a sign in Bengaluru : "Base Fare Rs 30, plus Rs 15 per km."
Can you write the magic math code for any distance?

Let k = kilometers $\rightarrow 30 + 15 \times k$



WHAT IS AN EXPRESSION?

An expression is a secret code waiting for inputs
— no equals sign needed!

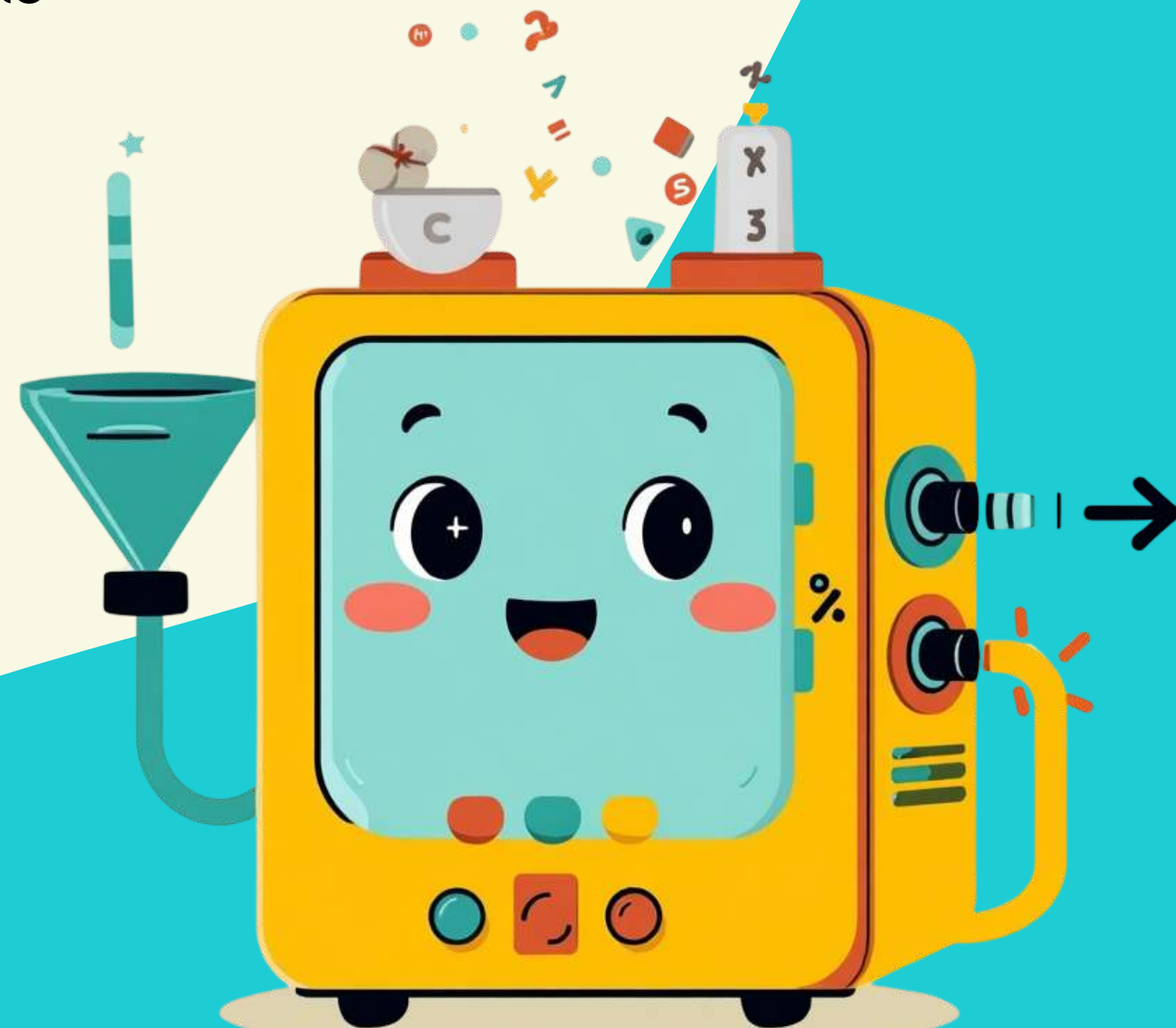
Metro card example:

Money loaded (m) minus Rs 5 fee

→ $m - 5$

● Variables (m)

● Constants (5)



EXPRESSION VS EQUATION

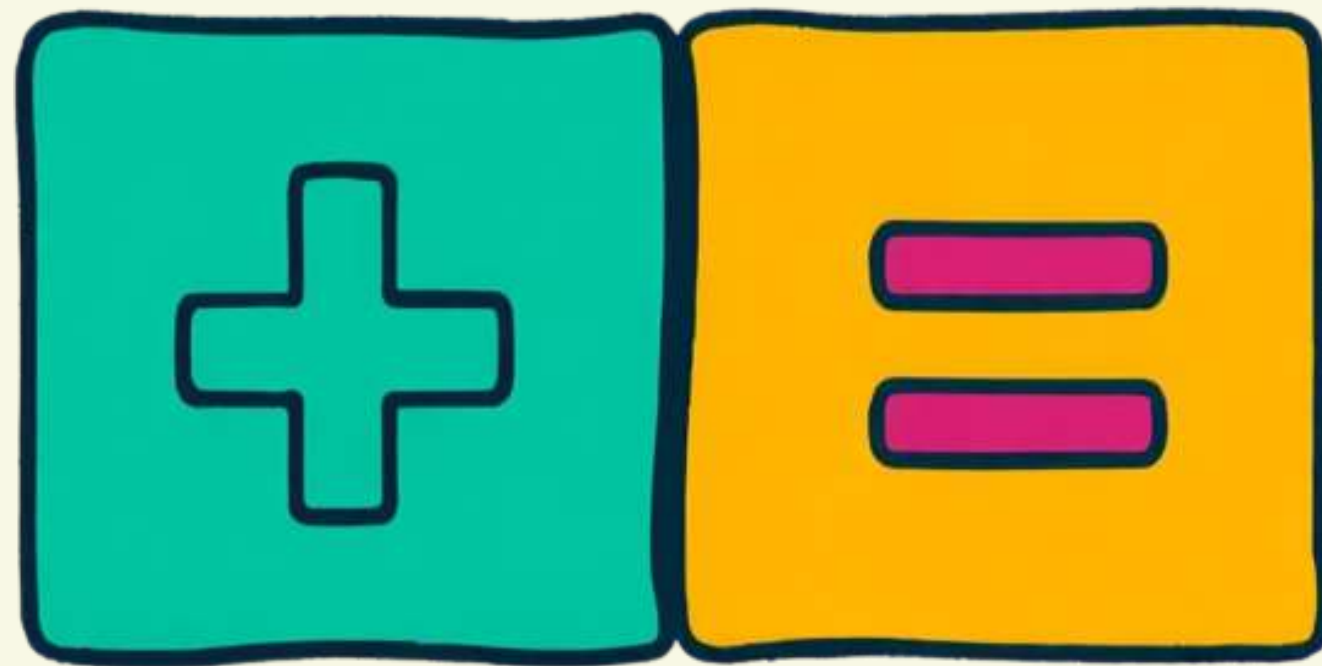
Expression: Code with NO equals sign

e.g., $x + 7$

Equation: Shows BALANCE with equals sign

e.g., $x + 7 = 12$

Spot the difference: the = sign makes it an equation!





TRANSLATION KEYWORDS: WORD TO MATH!

More than / Added to → Addition (+)

Less than / Taken away → Subtraction (-)

Times / Doubled / Tripled → Multiplication (×)

Shared / Divided among → Division (÷)

TRY IT! TRANSLATE THESE WORDS

Turn these word phrases into math expressions:

1. Seven more than a number (n) $\rightarrow$?
2. A number (p) decreased by four $\rightarrow$?
3. Triple a number (t) $\rightarrow$?
4. Twenty shared equally among friends (f) $\rightarrow$?

Answers :

1. $n + 7$

2. $p - 4$

3. $3 \times t$

4. $20 \div f$



WRITING EXPRESSIONS: 3 EASY STEPS

STEP 1

Find the unknown — what changes? That's your variable!

STEP 2

Spot the action word — is it more, less, times, or shared?

STEP 3

Write the code — put variable, operator & constant together!

Example: "Priya has 8 more stickers than Rohan" $\rightarrow r + 8$

THE SUBTRACTION TRAP!

Watch out! "5 less than a number" is NOT $5 - n$
It means: $n - 5$ (take 5 away FROM the number)

MORE EXAMPLES:

"8 less than m " $\rightarrow m - 8$ (not $8 - m$)
"Subtract 3 from y " $\rightarrow y - 3$ (not $3 - y$)
"Take away 10 from total t " $\rightarrow t - 10$

Remember: The number LOSES something, so it comes FIRST!



SOLVING EXPRESSIONS: PLUG IT IN!

Step 1: Write the expression

Step 2: Replace the variable with the given number

Step 3: Calculate the answer!

Example: Autorickshaw fare

Expression: $30 + 15 \times k$

If Ishaan travels 4 km, plug in $k = 4$:

$$30 + 15 \times 4 = 30 + 60 = \text{Rs } 90$$

Ishaan pays Rs 90 for his ride!

Try: If $k = 7$, what is the fare?

REAL-LIFE PRACTICE PROBLEMS

Clay Diyas

Priya makes d diyas each day.
She already has 12 diyas.
Total diyas after d days?

Expression: $12 + d$

Barfi Boxes

Each box has b barfis.
Ananya buys 3 boxes.
Total barfis?

Expression: $3 \times b$

Marigold Strings

Raj has m marigolds.
He gives 8 to his neighbour.
How many left?

Expression: $m - 8$

Now solve each one! If $d = 5$, $b = 6$, and $m = 20$,
what are the answers?

CHALLENGE ROUND!

Challenge 1: Translate AND solve!

"Aarav has some stickers (s). He gets triple, then gives away 4."
Write the expression. If $s = 10$, what is the answer?

Challenge 2: Spot the mistake!

"7 less than p " was written as $7 - p$.
Is this correct? Fix it if not!

Challenge 3: Create your own!

Write a word problem from your life. Translate it into an expression.

RECAP & TAKEAWAYS

Today We Learned:

- ✓ Expressions are math codes with variables, constants & operators
- ✓ Keywords help us translate words into math
- ✓ Watch out for the subtraction trap!
- ✓ Plug in values to solve expressions
- ✓ Math is everywhere - autorickshaws, diyas, barfis!

YOU ARE NOW EXPRESSION EXPERTS! ✨